Grade 6 Unit 8 Assessment Guide

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<tr><td>General Information</td><td>
The variety of questions that are possible for the benchmarks that are assessed in Unit 8 could not be covered in their entirety. To avoid a lengthy assessment the number of real-world questions were limited. The assessment includes questions where students can show their conceptual understanding of the formula for the area of a triangle and their understanding of packing cubes into a rectangular prism.

It is suggested that teachers monitor student understanding by noting if a student struggles with finding the area of a triangle, rectangle, or a quadrilateral. Teachers can use both the questions on area and on surface area to note any student weaknesses. Although students are allowed access to a calculator, teachers may also want to monitor if students are more successful on finding area when given whole numbers as in question 6 as opposed to rational numbers as in questions 4 and 5. Noting what types of barriers exist for your students can help you support their learning.

Students may need guidance in how to show their work when they have a calculator. Giving students an example such as the one shown may help students organize their work while also helping in the grading process. Equipping students to self-scaffold a problem through their work is a valuable outcome to requiring students to show their work.

work
area of triangle: $\frac{1}{2}(16.3)(22.4) = 182.56$
area of triangle: $\frac{1}{2}(12.6)(30) = 189$
area of rectangle: $(21.5)(16.3) = 350.45$
Total: $182.56 + 189 + 350.45 = 722.01$

Teachers may wonder why the units are given in the answer boxes. This is done so that a student cannot claim that they used a different unit to find the answer. There is not benchmark or MTR that specifies that a student must show knowledge of the correct use of a unit. Although this is an important distinction students should make, it is not assessed on this assessment. If teachers prefer to incorporate this into their assessment, then it is suggested that the teacher makes the necessary edits to the questions, the answer boxes, and the point distributions. Teachers are cautioned from over-emphasizing units when distributing points.
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<tr><td>Calculator Information</td><td>The questions on the unit assessment are calculator.</td></tr>
<tr><td>Total Recommended Points</td><td>The total recommended points in this guide are 100 points.</td></tr>
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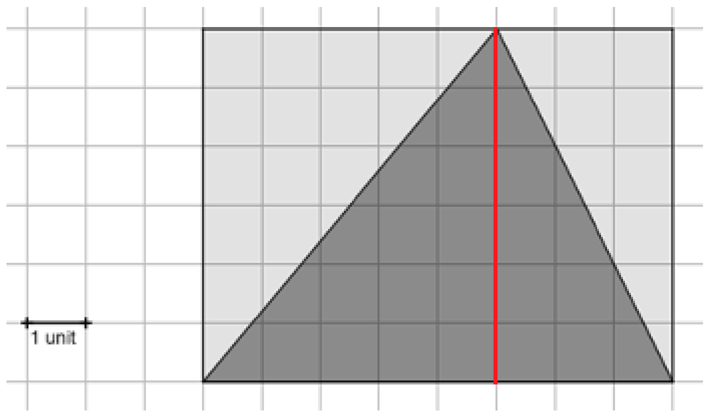
Question	1a	Benchmark(s)	MA.6.GR.2.1
		**Recommended Scoring** **(4 Total Points)**	Consider no partial credit.
		Additional Notes	Students could indicate the height in multiple locations using a vertical line. The answer shown in the key is not the only acceptable response to show the height.
Question	**1b**	**Benchmark(s)**	MA.6.GR.2.1
24 square units		**Recommended Scoring** **(5 Total Points)**	Consider no partial credit.
Question	**1c**	**Benchmark(s)**	MA.6.GR.2.1
c. Complete the statements. The triangle and the rectangle have o only the same base. o only the same height. o both the same base and the same height. The area of o all triangles is o any acute triangle is o any equilateral triangle is o any triangle that is inscribed in a rectangle is equal to o bh , o $b + h$, o $\frac{1}{2}bh$, o $\frac{1}{2}b + \frac{1}{2}h$, where b represents the base of the triangle and h represents the height of the triangle. **Students who select, “all triangles is” for the second answer choice should also be marked correct.*		**Recommended Scoring** **(10 Total Points)**	Consider giving students 2 points for each correct answer choice.
		Additional Notes	Both responses shown should be accepted. While the benchmark calls for students to connect their knowledge of the area of a rectangle to derive a formula for the area of a triangle, which is typically done with a triangle inscribed in a rectangle, a student who connects this to being true for all triangles should not be penalized.

<table>
<tr><td>Question</td><td>2</td><td>Benchmark(s)</td><td>MA.6.GR.2.1</td></tr>
<tr><td colspan="2">716.8 square feet</td><td>Recommended Scoring

(5 Total Points)</td><td>Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.</td></tr>
<tr><td>Question</td><td>3</td><td>Benchmark(s)</td><td>MA.6.GR.2.3</td></tr>
<tr><td colspan="2">4.25 inches</td><td>Recommended Scoring

(5 Total Points)</td><td>Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.</td></tr>
<tr><td>Question</td><td>4</td><td>Benchmark(s)</td><td>MA.6.GR.2.2</td></tr>
<tr><td colspan="2" rowspan="2">$1862\frac{187}{200}$ or 1862.935 square cm</td><td>Recommended Scoring

(20 Total Points)</td><td>Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider breaking the points up as follows:
• area of rectangle
• area of left triangle
• area of bottom triangle
• total surface area</td></tr>
<tr><td>Additional Notes</td><td>Look for students who struggle determining the dimensions of the bottom triangle. If students incorrectly determine the area of the bottom triangle but correctly add the area that they determined to find the total surface area, students should only lose the 5 points for the area of the bottom triangle.</td></tr>
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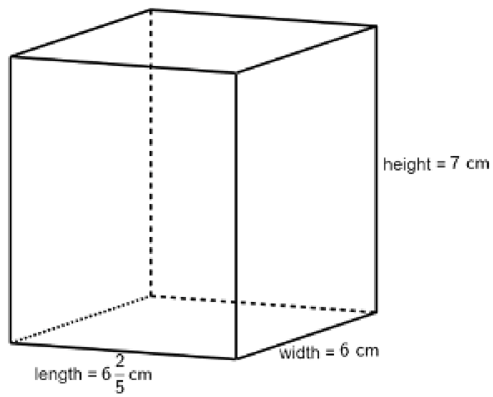
Question	**5**	**Benchmark(s)**	MA.6.GR.2.2
75 yards		**Recommended Scoring** **(5 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors.
Question	**6**	**Benchmark(s)**	MA.6.GR.2.4
340,928 square millimeters		**Recommended Scoring** **(20 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Consider breaking the points up as follows: • area of rectangle • area of left or right triangle • area of top or bottom triangle • total Surface Area

Question	7a	Benchmark(s)	MA.6.GR.2.3
A rectangular prism with dimensions in centimeters (cm) is shown.  a. Complete the statements. To find the volume of the prism, cubes with side lengths of ○ $\frac{1}{2}$ cm ○ $\frac{1}{5}$ cm ○ $\frac{1}{6}$ cm ○ $\frac{1}{7}$ cm can be used so that the length will have 32 cubes, the width will have 30 cubes, and the height will have 35 cubes. This results in 33,600 cubes, each with a volume of ○ $\frac{1}{8}$ ○ $\frac{1}{125}$ ○ $\frac{1}{216}$ ○ $\frac{1}{343}$ cubic centimeters.		**Recommended Scoring** **(21 Total Points)**	Consider giving students 3.5 points for each correct answer choice.
Question	**7b**	**Benchmark(s)**	MA.6.GR.2.3
$268\frac{4}{5}$ or 268.8 cubic centimeters		**Recommended Scoring** **(5 Total Points)**	Consider following student work to determine if a minor mathematical error was made. Consider partial credit for minor errors. Students may use cubes or the formula to determine the volume.